Year 9 Science Test 1: mark scheme Core paper

Variation and inheritance · Adaptation, extinction and natural selection · Cells and the microscope · Atoms and the periodic table · 40 minutes · 40 marks

Give the marks shown. Accept any correct equivalent wording. Where a part says "all four correct = 2 marks", do not award half marks.

Section A: Core

1 Environmental variation (5 marks)

Part	Answer	Marks
(a)	temperature: abiotic; predation: biotic; amount of light: abiotic; infectious disease: biotic. All four correct = 2 marks; two or three correct = 1 mark.	2
(b)	environmental	1
(c)	height: continuous; blood group: discontinuous; hand span: continuous; tongue rolling: discontinuous. All four correct = 2 marks; two or three correct = 1 mark.	2

2 Inherited variation (5 marks)

Part	Answer	Marks
(a)	nucleus (1), gametes (1), fertilisation (1), zygote (1).	4
(b)	natural eye colour and blood group, both ticked and no others, for 1 mark.	1

3 Adaptation and extinction (5 marks)

Part	Answer	Marks
(a)	A characteristic (feature) that helps an organism survive in its habitat.	1
(b)	nature reserve: an area where the habitat and the species living in it are protected (1); breeding programme: endangered animals are bred in captivity (1); gene bank: seeds, gametes or cells stored at a low temperature (1).	3
(c)	Any one of: greys can eat unripe acorns, so they eat the food before the reds do; greys store more body fat, so more survive the winter; greys carry squirrel pox, a disease that kills reds.	1

4 Natural selection and evolution (5 marks)

Part	Answer	Marks
(a)	genes passed on from parents	1
(b)	Correct order: a few moths had genes that made them black (1); soot made the tree trunks black (2); birds ate more pale moths (3); each generation had more black moths (4). All four correct = 2 marks; first and last correct = 1 mark.	2
(c)	A change over time in the characteristics of organisms (accept: a change in a species over many generations).	1
(d)	a theory	1

5 Cells and the microscope (5 marks)

Part	Answer	Marks
(a)	A cell wall (1), B nucleus (1), C chloroplast (1), D vacuole (1).	4
(b)	x40 (10 x 4)	1

6 Atoms and the periodic table (5 marks)

Part	Answer	Marks
(a)	neutron: 0, in the nucleus; electron: -1, in shells around the nucleus. All four correct = 2 marks; two or three correct = 1 mark.	2
(b)	(i) 11 protons (1). (ii) 23 - 11 = 12 neutrons (1).	2
(c)	2,8,1 (the missing number is 1).	1

Section B: Stretch and Challenge

7 Natural selection in the Salton Sea STRETCH (3 marks)

Part	Answer	Marks
(a)	By chance, some tilapia had genes (inherited variation) that helped them cope with salt (1). As the water got saltier, more of these fish survived and reproduced than the others (1), so they passed on their genes and each generation had more salt-tolerant fish, until all of them could cope (1). Do not credit 'the fish changed so they could cope'.	3

8 The size of a cell STRETCH (3 marks)

Part	Answer	Marks
(a)	(i) 10 x 40 = x400 (1). (ii) actual size = image size / magnification = 20 / 400 (1) = 0.05 mm (1). Accept 50 micrometres.	3

9 Groups and periods CHALLENGE (4 marks)

Part	Answer	Marks
(a)	(i) fluorine 2,7 (1); chlorine 2,8,7 (1). (ii) Both have seven electrons in the outer shell, and the outer electrons decide how an element reacts (1). (iii) Potassium is 2,8,8,1: its electrons fill four shells, and the period number is the number of shells (1).	4

Total: 40 marks